

CrackID: A Within-Visit Benchmark for Crack Re-Identification, and the Limits of Appearance Matching

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Abstract—Ticket management systems turn photographs into tracked defect records: a re-photographed crack must match its record or be filed as new. This work, developed within R2UTechnologies, studies within-visit crack re-identification under viewpoint and scale change. We benchmark ten zero-shot crop matchers from four families on CRACKID, a within-visit wall-crack benchmark. Ranking separates them (Rank-1 0.370–0.613; chance 0.336), but on the thresholded same-crack-or-not decision that a ticket record requires, all remain within $+0.037$ of the 0.337 chance floor. Adding wall context raises OSNet’s open-set rate from 0.168 to 0.404, yet erasing the crack retains 0.393: context retrieves on location, not crack appearance. A geometric reference that registers on wall texture with cracks masked and Chamfer-matches projected masks reaches an open-set rate of 0.768 at 48% coverage, against 0.384 for the best crop method; a coverage-only control scores 0.000. All accuracy rows are point estimates with no wall-level intervals. Ordering a gallery and deciding an identity are different capabilities, and the cue that works is location.

Index Terms—crack re-identification, dataset, benchmark, ticket management, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Ticket management systems convert photographs into tracked defect records: a re-photographed defect must eventually be *re-identified*, matched to its record, or given a new one. This work was developed in the context of ticket management within R2UTechnologies, where each defect report opens an issue ticket that later photographs must be linked to. This paper first studies within-visit re-identification under viewpoint change. Wall cracks are a common and difficult case, because the same crack is re-photographed from different distances, angles, and lighting, and grows between visits.

Crack analysis is a mature *detection* and *segmentation* task, and its datasets, CrackSeg9k [1] and OmniCrack30k [2], carry no cross-observation identity labels, because the task they serve needs none. We know of no benchmark for crack *re-identification*; the natural approach, crop-and-match as in person and vehicle re-ID [3], [4], assumes a crack has stable, locally distinctive appearance, and whether it does has never been measured. We introduce CRACKID, 140 photographs, 17 walls, 212 crack identities, to make that re-identification measurement (Fig. 2). Capture is one continuous walk-around per wall, so what is measured is *within-visit* viewpoint and scale correspondence, a hypothesized upper bound on multi-

visit performance, not a measurement of it; we add a synthetic revisit protocol as a controlled secondary check (Sec. VII-D). That check uses a distinct HYBRID-SKEL pipeline: it extends the Table II REG+CHAMFER pure registration-plus-Chamfer baseline with a skeleton-comparison fallback and per-pair blending ($\tau=8.0$ px), so its reversal does not directly contradict Table II.

Appearance buys almost nothing on the decision. On ranking the ten zero-shot matchers separate (Rank-1 0.370 to 0.613; chance 0.336 ± 0.025); on pairwise F_1 , the *same-crack-or-not* decision a record makes at a threshold, new rows land between $+0.000$ and $+0.037$ above the 0.337 best-constant floor (F_1^* 0.337–0.374), and at a validation-calibrated threshold all nine new rows sit at or below the floor (0.331–0.338). With no wall-level intervals, no fine ranking is claimed.

Two controls say where the difficulty is not (prior $n_Q=280$ snapshot, pending rerun). Giving an embedder the surrounding wall raises OSNet’s open-set rate from 0.168 to 0.404; *erasing the crack* from that view costs almost none of it (0.393). Such a method retrieves on the wall, and the cue that works, even implicitly, is *location*.

A geometric baseline makes location explicit (prior snapshot, not numerically comparable to the rerun). Registering on non-crack texture, projecting crack masks into a common frame, and Chamfer-matching under global assignment reaches DIR@FAR=0.1 of 0.768 against 0.384 (best new crop, SuperGlue) and 0.404 (best prior context), answering 48% of pairs; a COVERAGE-ONLY control scores 0.000 on the same metric, so the open-set gap is matching, not alignment coverage. This is a conditional, provenance-limited result, not a deployment-ready system: two test walls have zero registration success, and the annotation and baseline both use homography-based alignment. The resulting geometric advantage is therefore a hypothesis about this benchmark, not an independently established identity effect.

The revisit check reverses the ranking story. The real data are one continuous walk-around per wall, minutes apart, not separate visits. Under the small-warp synthetic protocol (affine $1.08\times$, 6° , 4° ; one seed; five walls), OSNet+context beats the HYBRID-SKEL at R@1 (0.826/0.671 versus 0.275/0.274) and open-set rate (0.739/0.663 versus 0.343/0.342) for growth/appearance change; the HYBRID-

SKEL’s small assignment- F_1 advantage is not evidence that geometry solves real multi-visit re-identification.

Contributions: (i) CRACKID with an explicit annotation-provenance report, 45% of identities formed by an automatic merge, reported rather than assumed (Sec. III-B); (ii) a ten-method zero-shot benchmark against stated chance levels, with crop+context and crack-erased controls (Sec. VI); (iii) a small single-seed synthetic revisit checkpoint isolating appearance change from damage evolution, the only multi-visit evidence this paper has (Sec. VII-D); (iv) a geometric reference baseline (REG+CHAMFER) with coverage-aware metrics, unanswerable queries, and an admission gate, kept distinct from the synthetic HYBRID-SKEL variant (Sec. V, IV, VII).

II. RELATED WORK

Crack analysis is overwhelmingly focused on detection and segmentation [5]–[7]; its datasets annotate cracks within single images and the large consolidations merging [1], [2] inherit that scope, and multi-view crack pipelines fuse several photographs of one inspection into a surface or a segmentation [8] without solving instance correspondence between them. Structural-health monitoring does compare cracks across time with machinery close to ours [9]–[12], but identity is always *given by construction* (fixed camera, marked location, operator at a viewpoint): the subject is one known crack, the question of width or extent, and no dataset supports method comparison on the association decision itself. Those systems therefore solve measurement after correspondence has already been supplied; they do not decide whether a newly observed crack is the same physical defect as a gallery crack under an open-set protocol. The benchmark targets that missing association decision, while its real capture scope remains within one walk-around rather than a true longitudinal study.

Object re-identification [4] with open-set identification [13] is the formal setting we adopt; keypoint matching [14], [15], learned correspondence (SuperGlue [16], LoFTR [17]), and region embeddings [3], [18]–[21] apply directly to crack crops, and our benchmark measures all of them zero-shot, without CrackID fine-tuning. We cite LightGlue [22] and place recognition (NetVLAD [23]) as the natural retrieval front end, but benchmark neither given compute and scope limits and the within-visit focus; they remain future baselines, not claimed comparisons. The YOLOv8 row uses the `yolov8n.pt` detector backbone (layer 9) as an embedding [24], version-pinned in code, not a fine-tuned crack model. Vehicle-damage assessment already re-identifies damage across claims in curated settings [25], a task the open-set protocol here mirrors; the rest of that literature treats damage detection and severity rating rather than association [26]–[28]. Whether the crop scope is itself the limiting choice for cracks is the question the benchmark is built to answer. The move the baseline makes, aligning two captures and then reasoning about what is present in one and absent in the other, is change detection [29], [30]; planar registration is classical [31]–[33] with photometric alignment (ECC [34]) as the fallback. The contribution here is not a new estimator but the observation that the alignment target

and the matching target should be *disjoint*: the homography is driven by the wall precisely because the crack is what is to be identified.

III. DATA ACQUISITION

A. Acquisition and composition

The benchmarked snapshot contains 140 photographs of 17 exterior walls acquired at one site over four half-days with two consumer handsets, at 3072×4080 . The walls span two surface families that behave very differently under keypoint matching: stippled render (densely textured) and smooth painted plaster, where the crack is frequently the only structure in frame. Walls split disjointly into 6 validation walls (66 photographs, 66 identities) and 11 test walls (74 photographs, 146 identities). Statistical power is governed by identities visible in two or more photographs, since only these form a query with an answer: 18 in validation, 19 in test. Each wall was captured in one continuous walk-around (consecutive frames a median of 3 s apart), so the dataset measures viewpoint and scale correspondence within one visit, a hypothesized upper bound on multi-visit performance, not yet a measurement of it.

Adjacent-frame retrieval. On the prior snapshot, 100% of 280 answerable test queries had their nearest correct gallery entry in the adjacent frame; on the current labels 95.8% of 306 do. Ranking therefore rewards short-baseline correspondence; Sec. VII sweeps the exclusion gap (prior pool $280 \rightarrow 205 \rightarrow 149 \rightarrow 105$; current pool $306 \rightarrow 247 \rightarrow 188 \rightarrow 124$ with floors $0.337 \rightarrow 0.288 \rightarrow 0.250 \rightarrow 0.204$) and reports that Rank-1 stays flat while the pool and prevalence change with it, so the sweep characterises robustness, not proof that duplicates are irrelevant.

B. Annotation, and why it needs auditing

One click point per crack is propagated across consecutive photographs through an ORB homography and one-to-one assignment; identities whose components appear to be fragments of one physical crack are then merged. The merge is where risk concentrates: a wrongly joined merge converts a hard negative into a free positive and inflates *every* method’s mAP together, so it cannot be caught by comparing methods. We therefore report its extent rather than assume its correctness: 95/212 identities (45%) were formed by a merge (48% on test), bridging a median maximum gap of 228 px; the adjusted Rand index against the pre-merge labeling is 0.080, meaning the merge restructured the partition rather than refining it. The triage is complete: every one of the 1,023 clicks no component claimed was reviewed against the imagery — 916 sit on visible cracks the segmenter missed rather than labelling mistakes, leaving 107 clicks (10.5%, 95% CI [8.7%, 12.5%]) naming no crack (dataset/labels/_unresolved.json). The 2,095 clicks resolve to 1,051 connected components; “points” below means clicks, not components. Single-annotator; agreement unmeasured, so the merge risk above is bounded by reporting, not removed. Counts in this section and Table I refer to the frozen benchmark snapshot (legacy directory

TABLE I

COUNTING UNITS FOR EVERY METRIC (PRIOR $n_Q=280$ SNAPSHOT; CURRENT TEST POOL: 306 ANSWERABLE, 30 390 VALID, 6 160 POSITIVE). PER SPLIT: WALLS, PHOTOGRAPHS AND IDENTITIES; IDENTITIES IN TWO OR MORE PHOTOGRAPHS (THE ONLY ONES THAT CAN FORM A QUERY WITH AN ANSWER); ANSWERABLE QUERIES (RANKING POOL); POSITIVE AND VALID QUERY-GALLERY PAIRS (PAIRWISE- F_1 POOL, PRIOR TEST PREVALENCE 0.180, CURRENT 0.203); AND SAME-WALL IMAGE PAIRS (REGISTRATION POOL, SEC. VII-B). THE REGISTRATION POOL (301 TEST PAIRS) DIFFERS FROM THE SCORED-PAIR SUBSET (255 PAIRS WITH VALIDITY-MASK CELLS ON THE PRIOR SNAPSHOT). COUNTS BELOW REFER TO THE PRIOR SNAPSHOT EXCEPT WHERE NOTED; SEE SEC. III-B FOR LABEL DRIFT.

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	280
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301

benchmark_out/, historic spelling retained for reproducibility); dataset/labels/ has since been re-edited (walls 18–19 added, audit in progress) and its raw identity/point totals no longer match exactly. The snapshot, its result.txt, chance.json, pair_outcomes.json, and viewpoint.json are the version of record for all accuracy rows.

IV. BENCHMARK PROTOCOL

The benchmark evaluates open-set retrieval: a photographed crack may or may not have a corresponding gallery instance. The protocol retains unknown cracks, calibrates thresholds on validation walls, and charges methods for unscorable pairs.

A *wall* is one physical wall at the capture site; all gallery candidates for a query come from that wall. An *identity* is a dataset label for a crack tracked across photographs, not an independently audited physical-identity judgment. A *query* is one annotated crack instance in a photograph. Its *gallery* contains instances from eligible photographs of the same wall, excluding the query photograph and capture session. A query is *known* if at least one valid gallery instance has the same identity; otherwise it is *unknown*. Test contains 306 answerable (known) and 199 unknown queries on the current labels (30 390 valid pairs, 6 160 positive, prevalence 0.203, floor 0.337; benchmark_out/chance_nq305.json). Table I retains the prior-pool recount for provenance; its test positives/valid counts (5 722/31 822) are superseded for the crop rerun. A *query-gallery pair* is one query compared with one eligible gallery instance; an *image pair* is two photographs of the same wall and is the unit charged for registration. Table I summarizes the pair and image-pair counts. On test, registration is attempted for all 301 same-wall image pairs; 255 contain at least one scored crack instance and produce validity-mask cells, while 46 yield no valid query-gallery pair after the 200 px size filter (see Sec. VII-B).

A. Metrics

Ranking uses Rank-1, Rank-5, and mAP over known queries. Pairwise F_1 classifies each valid query-gallery pair. Assignment F_1 (F_1^a in Table III) evaluates the one-to-one matches made independently for each image pair. We distinguish F_1^* (oracle threshold on test, a ceiling no deployment can pick), $F_1^{@v}$ (threshold calibrated on held-out validation walls, the deployable number), and F_1^a (assignment at the validation threshold). At threshold t , $\text{FAR}(t)$ is the fraction of unknown queries whose highest finite gallery score is at least t ; $\text{DIR}(t)$ is the fraction of known queries whose highest-scoring gallery instance is both correct and at least t . The reported $\text{DIR@FAR}=0.1$ is the largest DIR among thresholds whose FAR is at most 0.1 (a conservative open-set operating point; unknown queries with no finite score count as correct rejections, known queries with no finite score count as misses).

For assignment F_1 , the Hungarian algorithm maximizes total score within each query-image/gallery-image block; assignments below the validation-chosen threshold are rejected. Correct assignments are true positives, incorrect accepted assignments are false positives, and unmatched known queries are false negatives. Thus $F_1 = 2TP/(2TP + FP + FN)$; unmatched unknown queries are correct rejections. Unscorable pairs count as predicted negative for pairwise and assignment F_1 and rank last for retrieval. A known query with no finite score is a miss, while an unknown query with no finite score is not a false alarm.

Every metric is reported against its chance level, because two are far from zero (current test pool, chance_nq305.json: 200 random matrices): pairwise F_1 has a best-constant-predictor floor $2p/(1+p) = 0.337$ at prevalence 0.203, Rank-5 chance 0.778 ± 0.019 ; Rank-1 0.336 ± 0.025 and DIR@FAR 0.038 ± 0.014 , assignment- F_1 0.345. Prior-pool values were 0.305/0.767/0.336/0.038. We call $2p/(1+p)$ a floor, not coin-flip chance: it is the score of predicting every pair positive, and it changes when the evaluated pair pool changes. Confidence resampling uses walls rather than queries because queries from one wall share capture conditions. With 11 test walls, no numeric interval is attached; wall-level intervals await a larger dataset and release of the per-pair score matrices (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The ceiling motivates a change of scope. A crack is a mark at a fixed location on a rigid, near-planar surface that carries texture the crack does not; we identify a crack by *where it is on its wall* once the two photographs are aligned. We release this as a deliberately simple *reference baseline* for CRACKID, with no learned component beyond the shared crack segmenter. The segmenter is a UNet++ model (segmentation-models-pytorch, batch_segment.py / tiled_inference.py) whose binary masks are filtered into instances with minimum area 200 px and morphological closing 5 px; the same masks and instance extraction serve every method, so recall failures affect

Same identity, frame 0001



Stippled render



Same identity, frame 0009



Smooth painted plaster



Fig. 1. Examples from real CRACKID captures. Top: the same annotated identity in non-adjacent frames of one walk-around. Bottom: the two surface families, stippled render and smooth painted plaster. The identity labels inherit the dataset’s homography-based propagation and are illustrative, not an independent annotation audit.

all rows identically. All baselines are zero-shot on CRACKID: no backbone is fine-tuned on crack pairs.

A. Alignment driven by non-crack texture

A wall is planar, so the homography $x' \sim Hx$ mapping photograph I_a into I_b ’s frame is the correct model. The design decision that matters is what drives the fit: we dilate the crack masks by 15px and exclude those pixels from detection, so H comes from wall texture alone. This is not an optimization but what makes the projected position an independent identity cue: were cracks to participate, a long crack could dominate the correspondence set and pull the fit toward the wrong crack. Detection runs on copies downsampled so the long side

is at most 1600px, after CLAHE equalisation (clip limit 2.0, 8×8 tiles, OpenCV). We use SIFT with up to 8000 features (OpenCV), FLANN matching with a 0.75 Lowe ratio test, and USAC_MAGSAC where available (falling back to RANSAC) at 4px with up to 10 000 iterations, requiring 25 inliers.

B. A cascade for two surface families

One front end does not serve both surfaces, and the failure is categorical: On the smooth-plaster walls, crack-masked SIFT succeeds on *exactly zero* pairs. We try three stages in order: (i) SIFT on masked wall texture, for stippled render; (ii) SIFT with cracks restored, triggered only when masking starves detection below 300 keypoints on the weaker side (on

plaster the crack is the only feature in frame), accepting the circularity stage (i) avoids, and recording which front end produced each result; (iii) ECC photometric alignment over a $320 \rightarrow 640 \rightarrow 1280$ px pyramid (translation then Euclidean initialisation, up to 300 iterations per level), accepted only if enhanced correlation exceeds 0.80. Every candidate H is sanity-checked (reject mirrored or folded quadrilaterals, area scale outside $[1/16, 16]$, collapsed sides); its full-resolution inlier RMS sets the Chamfer tolerance for that pair (Sec. VII-C). Stage identities (`sift`, `sift-nocrackmask`, `ecc`) are logged per image pair.

C. Projection and Chamfer agreement

Crack instances are connected components; pixels from I_a are projected by H and compared with instances of I_b . Chamfer distance, not IoU, is the criterion: cracks are a few pixels wide, so a 3 px registration error drives IoU to zero for a correctly aligned pair. Let $d(p, b) = \min_{q \in b} \|p - q\|_2$ be the point-to-set Euclidean distance and $|a|$ the pixel count of mask a . The reported score is the **symmetric mean Chamfer distance** in full-resolution pixels

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

over the union bounding box, with cheap rejection when boxes are far apart. We also report a scale-free alternative, **Chamfer coverage**, the symmetric fraction of pixels within τ , $\frac{1}{2}(|\{p \in a : d(p, b) \leq \tau\}|/|a| + |\{q \in b : d(q, a) \leq \tau\}|/|b|)$, with τ set per pair from that pair's own inlier RMS, clamped to $[2, 16]$ px (default 6.0 px when RMS is unavailable; the synthetic HYBRID-SKEL variant fixes $\tau=8.0$ px). Eq. (1) is a per-pair pixel distance whose scale differs per pair, so a single global threshold makes ranking partly measure calibration (Sec. VII-C).

D. Explicit non-answers, and the licensed negative

Correspondence is a one-to-one assignment per image pair with an explicit rejection threshold, not a per-query argmax: two cracks must not both claim the same crack; unmatched instances are reported as *disappeared* or *newly appeared*. The Hungarian solve [35] is part of *evaluation*, applied identically to every method's score matrix. Specific to this formulation is the evidence a **licensed negative** can carry: with I_a projected into I_b 's frame, the system observes a region visible and empty of the recorded crack, positive evidence of absence, categorically stronger than the low similarity a crop embedding produces. A projected crack falling outside the gallery photograph is a confident rejection, counted separately from unscorable pairs; this is the mechanism behind the open-set gap (Sec. VII-B). Unscorable pairs (no registration, or no retained instance after the 200 px size filter) receive $-\infty$ and are charged as predicted negative for F_1 and rank last for retrieval.

VI. THE APPEARANCE CEILING

A. Setup

Ten crop-scope matchers run under an identical protocol, identical inputs, and identical scoring: handcrafted keypoints (SIFT [14], ORB [15]), learned correspondence (SuperGlue [16] indoor weights, LoFTR [17] indoor weights, both at 512 px long side), general embeddings (ViT-B/16 [18], DeiT-S [19], CLIP ViT-B/32 [20] `openai/clip-vit-base-patch32`, DINOv2 [21] ViT-S/14, YOLOv8 [24] `yolov8n.pt` layer 9), and a re-identification architecture (OSNet [3] `osnet_x1_0` default weights). All image backbones are zero-shot on CRACKID with ImageNet or web pretraining only; no backbone is fine-tuned on cracks. Embedding crops are letterboxed to 224×224 with ImageNet normalisation and scored by cosine similarity; keypoint and learned correspondence scores are RANSAC inlier counts under the same validity mask. LightGlue [22] and NetVLAD [23] are cited as natural stronger retrieval front-ends but are not benchmarked here; they remain future baselines given compute and scope limits.

B. Result: appearance buys almost nothing on the decision

On the decision a record actually makes, same crack or not, at a threshold, no new crop row clears the best-constant floor by more than 0.037 (excess over the 0.337 floor at test prevalence 0.203, Table II: new F_1^* 0.337 to 0.374, i.e. $+0.000$ to $+0.037$; three rows at the floor to three decimals), spanning handcrafted keypoints to foundation-model embeddings and more than two orders of relative compute (0.006 s to 7.19 s per query in separate per-method runs, Table II; absolute timings are workstation-dependent). At a deployable (validation-calibrated) threshold, the nine new rows sit at $F_1^{\text{@}v}=0.331-0.338$ against the 0.337 floor (-0.006 to $+0.001$): thresholded decisions are at or below chance. Ranking does not transfer to deciding: the nine new rows separate over a 0.243 range on Rank-1 (0.370–0.613; chance 0.336 ± 0.025), yet ViT takes best Rank-1 (0.613) while its oracle pairwise F_1^* sits only 0.007 above the floor, and SuperGlue (seventh on Rank-1) has the best new crop open-set rate (0.384; chance 0.038 ± 0.014). Ordering a gallery well and keeping a record are different capabilities. The collapse after validation calibration is consistent with threshold instability under a small, wall-clustered sample: methods can order candidates differently while their score distributions do not support a stable same-crack/not-same-crack boundary on held-out walls. This is an interpretation of the observed pattern, not a claim that calibration itself is the sole cause. With 19 matchable test identities and no wall-level intervals, no fine ranking among crop baselines is claimed.

One contributor is checkable from the counting units. Validation pair prevalence was 0.292 (26 012/89 118) on the prior snapshot against 0.203 on the current test pool (6 160/30 390), so the F_1 -optimal threshold is not a property of a method but of the pool it was fitted on: the same scores that separate cracks on the denser validation pool place their boundary in the wrong place once prevalence shifts, and every method shifts

TABLE II

MAIN RESULT, TEST SPLIT, $n_Q = 305$ QUERY INSTANCES OVER 11 WALLS. ALL METHODS SEE IDENTICAL INPUTS AND ARE SCORED BY IDENTICAL CODE. F_1^* IS ORACLE-THRESHOLDED ON TEST (CEILING, NOT DEPLOYABLE); ΔF_1^* IS ITS EXCESS OVER THE 0.337 BEST-CONSTANT-PREDICTOR FLOOR ($2p/(1+p)$) AT TEST PREVALENCE 0.203, NOT COIN-FLIP CHANCE; SEC. IV); $F_1^{\text{@v}}$ IS THRESHOLDED ON THE HELD-OUT VALIDATION WALLS AND IS THE DEPLOYABLE NUMBER. *scored* IS THE FRACTION OF VALID QUERY-GALLERY PAIRS RECEIVING A FINITE SCORE; UNSCORED PAIRS ARE CHARGED AS PREDICTED-NEGATIVE, AND THE QUERY IS STILL COUNTED (SEC. VII-B). COST IS RELATIVE WALL-CLOCK SECONDS PER QUERY (NEW CROP ROWS: TOTAL_s/305 FROM SEPARATE PER-METHOD RUNS; USE FOR ORDERING, NOT ABSOLUTE TIMING). CHANCE LEVELS ON THE CURRENT LABEL POOL (BENCHMARK_OUT/CHANCE_NQ305.JSON: RANK-1 0.336 ± 0.025 , RANK-5 0.778 ± 0.019 , MAP 0.382, DIR@FAR 0.038 ± 0.014 , PAIRWISE- F_1 FLOOR 0.337, ASSIGNMENT- F_1 0.345; 200 RANDOM MATRICES). BOLD MARKS THE BEST CROP-SCOPE VALUE PER COLUMN ONLY (NEW ROWS ONLY; † CLIP-CROP, COVERAGE-ONLY, AND REG+CHAMFER ROWS ARE THE PRIOR $n_Q=280$ SNAPSHOT PENDING RERUN AND MUST NOT BE COMPARED NUMERICALLY TO THE NEW ROWS. THE COVERAGE-ONLY REFERENCE ROW ANSWERS YES TO EVERY REGISTERED PAIR WITHOUT EXAMINING A CRACK AND IS THE LEVEL THE GEOMETRY ROW MUST BE READ AGAINST. COUNTS REFER TO THE CURRENT LABELS; DATASET/LABELS/ REMAINS UNDER AUDIT (SEE SEC. III-B).

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{\text{@v}}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope (new $n_Q=305$ rerun except CLIP)</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.000	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.006	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.020	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.020	0.337	1.00	7.190
YOLOv8 backbone [24]	embedding	0.525	0.439	0.134	0.374	0.037	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.007	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.000	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding †	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.000	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.001	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	0.351	0.582	0.48	27.92

together. The current test floor is 0.337 (validation floor TBD after val-split rerun with walls 18–19). This is a mechanism, not a measurement: the threshold-free version of the claim (pairwise average precision, or AUROC against prevalence) would settle it, and we do not report it; releasing the saved score matrices it requires is listed as future work (Sec. VIII).

C. Is it the crack, or the wall behind it?

Two controls separate identity cue from field of view at fixed backbone: a *crop+context* condition (@ctx1: instance bounding box expanded by $1.0\times$ its size on each side) and a *crack-erased* condition (@ctx1@nocrack: the same widened view with the crack mask dilated by 5px and filled by Telea inpainting, radius 3px, so the silhouette is removed rather than blacked out). Context helps a lot: OSNet’s Rank-1 rises from 0.654 to 0.775, mAP to 0.595, open-set rate from 0.168 to 0.404, and its pairwise F_1^* from 0.309 to 0.354 (prior $n_Q=280$ snapshot; context controls pending rerun at $n_Q=305$); CLIP’s open-set rate rises from 0.143 to 0.311, but erasing the crack costs almost none of it: OSNet retains Rank-1 0.711, mAP 0.574 and open-set 0.393; CLIP retains Rank-1 0.600, mAP 0.502, open-set 0.286. Both erased views still outrank every tight-crop method: the retrieved cue is the wall, that is, location. Pairwise F_1 moves little with context on the prior pool: all four context variants lie between 0.346 and 0.361 (prior floor 0.305), over a change that gives OSNet +0.236 of open-set rate. We read the aggregate as a property of the data, pending rerun at the current pool: methods with

no shared inductive bias land near the floor, and wherever a crop method improves, the improvement is not the crack. Ten methods are a sample, not proof; the result establishes that the obvious moves buy at most 0.037 above the 0.337 floor on the decision under the new rerun, reason to change the formulation, not the encoder.

VII. RESULTS

A. Main comparison and the open-set gap

On different pair pools, the geometric reference (prior $n_Q=280$ snapshot) changes the operating regime and is not numerically comparable to the new $n_Q=305$ crop rerun: prior crop Rank-1 0.654 (best) against geometric Rank-1 0.952, crop mAP 0.520 against 0.886, but the pools differ (*scored* 1.00 vs 0.48). Conditioned on coverage, as excess over the COVERAGE-ONLY reference, the prior ranking gains are +0.437/+0.353, while prior pairwise $F_1^* = 0.656$ is +0.112 over 0.544, not the +0.351 a comparison against the crop floor would suggest. The open-set result is the one number coverage does not explain: prior DIR@FAR=0.1 reaches 0.768 against 0.384 for the best new crop method (SuperGlue at $n_Q=305$), 0.404 for the best prior context control, chance 0.038 ± 0.014 , and exactly 0.000 for COVERAGE-ONLY. Observed: crop methods return a similarity to the nearest neighbor, which on self-similar hairline cracks is often high, so they cannot decline. Interpretation: an aligned pair supports the *licensed negative*: the query projects into the gallery frame, where its region is visible and empty. Relative compute does not explain

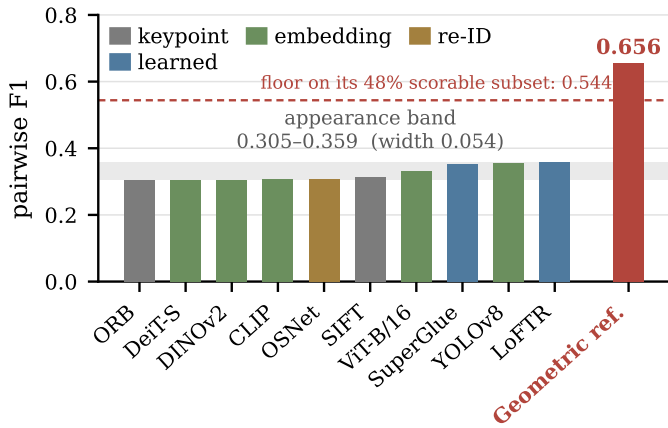


Fig. 2. The appearance ceiling. Pairwise F_1 for ten crop-scope matchers spanning four method families on CRACKID test ($n_Q=305$ for nine rows; CLIP-crop prior snapshot). New rows fall inside 0.337 to 0.374 against the 0.337 best-constant floor ($2p/(1+p)$ at prevalence 0.203), irrespective of family or relative compute. The geometric reference baseline (red, prior snapshot) leaves the band, but it answers only 48% of pairs, for which the best-constant floor on the current labels is 0.558 (the COVERAGE-ONLY level with old registration outcomes), not 0.337: its bar is drawn against the dashed line and must not be read against the crop floor. Ranking metrics separate the same methods over a 0.24 range; the decision metric does not. All bars are point estimates on the rerun pool with no wall-level intervals. Figure bars show the prior-pool band and await regeneration after the full-pool rerun; numbers in Table II take precedence.

the gap: LoFTR is the costliest new crop row (7.19s/query), OSNet the cheapest (0.006s/query) in separate per-method runs (Table II; absolute timings workstation-dependent).

B. Partial coverage, gap structure, and failures

Every crop method scores 100% of valid pairs, the geometric method 48%; Unscorable pairs are charged as predicted negatives. For ranking that is a penalty: censoring a complete matrix drops Rank-1 0.422 to 0.27 at 34% coverage in a held-out control (score matrices not committed; see paper README). For pairwise F_1 it is a subsidy: the scorable subset retains 93.0% of positives and 37.5% of negatives on the current labels with old registration outcomes (100%/36.8% prior), so prevalence rises 0.203 to 0.387 and the floor with it, 0.337 to 0.558 (prior: 0.180→0.373, 0.305→0.544). COVERAGE-ONLY scores 0.544 pairwise F_1 on the prior pool (0.558 on current labels with old outcomes) and mAP 0.533 on the prior pool (both above prior OSNet 0.309/0.520; current-pool rerun of coverage/geometry pending), and Rank-1 0.515 prior-pool, by answering every registered pair without examining a crack. Its signal is registration overlap: it assigns one constant score to registered image pairs and $-\infty$ to unregistered pairs, so candidate images are ranked even though crack pixels are never inspected. Much of the ranking and pairwise- F_1 margin is therefore *which pairs align*; the open-set metric is the exception and the result we lead with.

Excluding the adjacent frame, then the two and three nearest, leaves Rank-1 flat on the prior pool (OSNet 0.654 to 0.648) with excess over chance stable in 0.24 to 0.32: the near-duplicate structure is real but the flat curve alone does

not prove it is irrelevant, because the answerable pool shrinks (prior 280→105; current 306→124) and prevalence shifts with it. It is also important for interpretation: the adjacent frame answers almost all queries (100% prior; 95.8% current), so the headline ranking measures short-baseline correspondence rather than broad temporal re-identification. Registration success is strongly wall-dependent (97/301 pairs, 32.2%: walls 07/08/12 100%, wall 09 52%, wall 11 83%, walls 13-16 between 3% and 29%, walls 10/17 at 0%). The pairs that register are the densely annotated ones, so the 96 registered pairs that carry a scored query-gallery pair hold 48% of the 31822 valid ones — 160 valid cells per image pair against 104 for a failed pair — from a third of the available pairs (46 of the 301 same-wall pairs yield no valid pair at all). A logistic fit over all 301 pairs makes the mechanism plain: sharpness dominates (median 133.8 registered vs 7.2 failed, wall-clustered Wald $z = 5.1$, $p = 3 \times 10^{-7}$), frame gap far less ($z = -2.8$, $p = 0.005$). The independent-pairs z s (8.2, -5.0) overstate the contrast because failure is categorical by wall, so we resample walls, not queries, and the leave-one-wall-out spread in Rank-1 is large, so no fine ranking among baselines is claimed. One admission gate (variance-of-Laplacian ≥ 10) applied identically rejects 21 of 74 test captures and nearly doubles registration, 32.2% to 56.1%.

C. Ablations and cost

Chamfer distance versus coverage. The scale-free coverage score motivated in Sec. V is not a free improvement: it drops mAP 0.886 to 0.646 and Rank-1 0.952 to 0.849, the largest movement any change produces; DIR and $F_1^{@v}$ barely move (0.768→0.754, 0.582→0.556), so coverage gives a threshold decision, not a ranking. Masking. Disabling crack exclusion leaves ranking and pairwise agreement practically unchanged (to three decimals) but raises DIR@FAR 0.768 to 0.825: masking is not what buys the reported accuracy. This is expected when the test set contains no deliberate case where crack pixels create a spurious registration: masking buys evidence independence, not observed accuracy on this collection. We retain masking as the primary configuration because it keeps the alignment target disjoint from the identity target, and report the nomask +0.057 openly as a safeguard awaiting a harder stress test. ECC fallback. With ECC enabled (primary): Rank-1 0.952, mAP 0.886, *scored* 0.48; with ECC disabled: Rank-1 0.920, mAP 0.862, *scored* 0.50. Assignment F_1 rises 0.769 to 0.784 when ECC is enabled; open-set DIR is unchanged. ECC was kept versioned after a fix for a configuration in which it had been inert.

Cost. The geometric pass costs 7817s total on test in our sessions (27.9s/query if charged per query, about 26s per same-wall image pair over 301 pairs). That per-query figure is a charging convention, not an intrinsic one: registration is per image pair, one homography serves every crack on a wall, and amortized over the 301 test image pairs the same pass runs about 26s/pair, after which any later crack query on the pair pays only the Chamfer pass, not the homography. Absolute timings are workstation- and load-dependent and are

Protocol	Method	n_q	R@1	mAP	DIR@FA	Fallback of blend, F_1^a from shared	$F_1^{@v}$	$H\%$
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646	1.00
A (growth)	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.640	1.00
B (appearance)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640	1.00
B (appearance)	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609	1.00

TABLE III

HYBRID-SKEL VERSUS THE STRONGEST APPEARANCE BASELINE ON THE SYNTHETIC REVISIT PROTOCOL (SEC. VII-D); PAIRWISE AND ASSIGNMENT F_1 AT THE VALIDATION-CALIBRATED THRESHOLD. ONE SEED (0), TEN SOURCE PHOTOGRAPHS ACROSS FIVE WALLS, NO RESAMPLING; ROWS CARRY NO UNCERTAINTY AND ARE AN ACCURACY CHECKPOINT, NOT A STABILITY CLAIM (SEC. VII-D). HYBRID-SKEL IS NOT THE TABLE II REG+CHAMFER CONFIGURATION.

reported for relative ordering only; relatively, the pair-level cost is within an order of magnitude of LoFTR, a crop method with none of the open-set benefit. No fine ranking by cost is claimed.

D. A synthetic revisit protocol

The operational setting a ticket management backend serves is a *revisited* crack — one that already has an open issue ticket — so we add a second protocol isolating controlled change. Ten source photographs across five of the walls (walls 01, 02, 04, 05, 09; a single seed, 0) are each synthesized into a set of queries under an identical affine warp ($1.08\times$ scale, 6° rotation, 4° tilt; affine approximation, not a full projective revisit): dataset A additionally grows a single tip of the largest crack component (length $0.18\times\max(16, W/14)$ px, width 3px, tip farthest from the component centroid, `hybrid_eval.py`), dataset B corrupts appearance while leaving structure untouched (Gaussian blur $\sigma=1.6$, noise $\sigma=14$, contrast/brightness shift); each synthetic query is ranked against the wall’s real instances only. Unlike the Table II REG+CHAMFER system, which is pure registration plus Chamfer scoring with per-pair RMS tolerance, the synthetic-protocol HYBRID-SKEL folds a structural skeleton comparison into the geometric baseline as a fallback plus per-pair blend ($\tau=8.0$ px), scored by the same code as OSNet+ctx (Table III). HYBRID-SKEL and REG+CHAMFER are therefore different systems; the synthetic rows test generalization under controlled change, not the Table II configuration. The warp is deliberately modest (real walk-around tilt reaches $p90\ 75^\circ$), so this is a checkpoint on small geometric change with controlled photometric or growth change, not a real revisit.

The geometric side does not transfer its main-benchmark advantage here: OSNet+ctx separates the synthetic queries at R@1 0.826/0.671, the HYBRID-SKEL at 0.275/0.274, with open-set rates 0.739/0.663 against 0.343/0.342. But the split is metric-dependent, and opposite the ranking gap: on identity agreement, the HYBRID-SKEL holds its own, with assignment $F_1^a = 0.646/0.640$ beating OSNet’s 0.631/0.609, and pairwise $F_1^{@v} = 0.569/0.574$ trailing 0.601/0.594 by far less than the ranking margin. Both rate 1.00 scored, and per assignment logs none was decided by the pure-homography branch alone ($H\% = 0.0$, the share of assignment cells whose verdict came straight from the homography with no skeleton

rather than Table III), the geometric pipeline answers *which crack is 0.601 on pairs appearance also resolves, but loses the global ranking of a hundred-plus same-wall candidates at one threshold*. These rows are a single-seed checkpoint on five walls, not a claim: they show the Table II advantage does not imply multi-visit robustness.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This benchmark establishes a first measurable formulation of crack re-identification; several directions remain open and represent natural extensions of the present work.

Scaling to multi-wall and multi-visit datasets. The test split covers tens of matchable identities across 11 walls (19 on the prior snapshot; recount pending on current labels), which is sufficient to establish the structural pattern (thresholded decisions at or below the floor; open-set benefit of explicit alignment) but not to rank individual methods with statistical confidence; all accuracy rows are point estimates with no wall-level intervals. A larger dataset — more walls, more surface families, and more annotators — would allow wall-level confidence intervals to be reported and method rankings to be interpreted reliably. More importantly, the current benchmark captures one continuous walk-around per wall; a second capture session at deliberately varied distances and lighting would convert the within-visit measurement into a genuine cross-visit benchmark. That real revisit data is the most valuable next step for assessing the re-identification component of operational ticket management backends.

Annotation quality and provenance-free labelling. Annotation quality was actively managed: a single human annotator placed and reviewed all 2,095 clicks from homography-propagated prefill until zero provisional points remained, and every one of the 1,023 clicks no component claimed was subsequently triaged against the imagery — 916 (89.5%) sit on visible cracks the segmenter missed rather than labelling mistakes, leaving 107 clicks (10.5%, 95% CI [8.7%, 12.5%]) naming no crack. Single-annotator; agreement unmeasured. The reported geometric advantage must therefore be read as provenance-conditioned until a small, manually verified subset labeled entirely without homography propagation quantifies how much is independent of annotation provenance. Releasing the per-pair score matrices alongside the dataset would additionally allow third parties to run sensitivity analyses on how much false-positive mass would be needed to manufacture an open-set gap of the observed magnitude, and to compute wall-clustered intervals and threshold-free AP/AUROC.

Closing the registration coverage gap. The geometric baseline currently answers 48% of query pairs and scores zero on smooth-plaster walls. Future work should explore surface-adaptive registration strategies — including learned local features, direct alignment on photometric gradients, or sparse depth from consumer devices — that could push coverage toward full-set evaluation. A learned front-end replacing

SIFT would also decouple the geometry pipeline from the homography-based annotation provenance.

Learning to re-identify from geometry-derived supervision.

The geometric baseline is a deliberate non-learned reference; it is not a deployment target. A natural next step is to use the registration pipeline as a teacher to generate pseudo-labels for training a learned matcher, or to fine-tune a foundation-model backbone on crack pairs with homography-derived correspondence. This would let the system inherit the location-based advantage while remaining applicable to pairs the registration pipeline cannot answer.

Integration into ticket lifecycle evaluation. End-to-end evaluation in a ticket management context requires measuring not just retrieval accuracy but downstream impact: how often does a mis-identification open a duplicate ticket, and how often does a false match suppress a new defect? Future work should define task-level metrics (duplicate rate, missed-new-defect rate) and evaluate the full pipeline — segmentation, re-identification, and ticket assignment — in a closed-loop simulation that reflects real inspection workflows.

IX. CONCLUSION

We built a benchmark that makes crack re-identification testable: CRACKID, 140 photographs over 17 walls with 212 identities in the frozen snapshot, its annotation provenance (45% merged) stated rather than assumed, with the triage of unclaimed clicks complete (107/1 023 naming no crack). All accuracy rows are point estimates with no wall-level intervals (19 matchable test identities on the prior snapshot); we claim the pattern, not the digits. The benchmark’s main finding is negative and structural: on the decision a record requires, ten zero-shot matchers across four families buy between +0.000 and +0.037 over the 0.337 best-constant floor (F_1^* 0.337–0.374), with all nine new rows at or below the floor at the validation threshold ($F_1^{@v}$ 0.331–0.338). Ordering a gallery and deciding an identity are not the same capability. The difficulty is not crack appearance alone: a geometric reference baseline (prior snapshot) that registers on wall texture with cracks masked out and Chamfer-matches projected masks reaches DIR@FAR=0.1 of 0.768 against 0.384 for the best new crop method and 0.404 for the best prior context method while answering 48% of pairs, and a COVERAGE-ONLY control exploiting that alignment without examining a crack scores 0.000 on the open-set metric (prior 0.544; 0.558 on current labels with old registration outcomes). Operationally, the open-set advantage comes from an aligned pair supporting a *licensed negative*, positive evidence of absence, where a crop embedding can only report low similarity; whether that advantage is independent of homography-based annotation remains unresolved and must be read as provenance-conditioned. Two bounds remain addressable: a real second capture session turns the benchmark from viewpoint to cross-visit, and a second annotator plus a homography-free audit subset would turn the single-annotator triage into a measured agreement rate. The 48% coverage is also a deployment gap in its own

right, not merely a scoring issue: smooth-plaster walls can be categorically unanswerable. Finally, the small single-seed synthetic revisit checkpoint reverses the headline ranking and open-set result in favour of OSNet+context, while the distinct HYBRID-SKEL pipeline retains only a narrow assignment- F_1 advantage; the present paper therefore establishes a promising within-visit geometric reference, not a solved re-identification component for a multi-visit ticket management backend.

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REFERENCES

- [1] S. Kulkarni, S. Singh, D. Balakrishnan, S. Sharma, S. Devunuri, and S. C. R. Koralapati, “CrackSeg9k: A collection and benchmark for crack segmentation datasets and frameworks,” *arXiv:2208.13054*, 2022.
- [2] C. Benz and V. Rodehorst, “OmniCrack30k: A benchmark for crack segmentation and the reasonable effectiveness of transfer learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.
- [3] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, “Omni-scale feature learning for person re-identification,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 3702–3712.
- [4] Y. Qian, J. Barthelemy, E. Karuppiah *et al.*, “Identifying re-identification challenges: Past, current and future trends,” *SN Computer Science*, vol. 5, p. 937, 2024.
- [5] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, “CrackTree: Automatic crack detection from pavement images,” *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.
- [6] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, “DeepCrack: A deep hierarchical feature learning architecture for crack segmentation,” *Neurocomputing*, vol. 338, pp. 139–153, 2019.
- [7] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sessmann, D. Ebersbach, U. Stoeckert, and H.-M. Gross, “How to get pavement distress detection ready for deep learning? a systematic approach,” in *Int. Joint Conf. Neural Networks (IJCNN)*, 2017, pp. 2039–2047.
- [8] C. Benz and V. Rodehorst, “CrackStructures and CrackEnsembles: The power of multi-view for 2.5d crack detection,” in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5990–5999.
- [9] L. Parente, E. Falvo, C. Castagnetti, F. Grassi, F. Mancini, P. Rossi, and A. Capra, “Image-based monitoring of cracks: Effectiveness analysis of an open-source machine learning-assisted procedure,” *Journal of Imaging*, vol. 8, no. 2, p. 22, 2022.
- [10] S. Harb, P. Achancaray, M. Maboudi, and M. Gerke, “Multi-temporal crack segmentation in concrete structures using deep learning approaches,” *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. X-G-2025, pp. 341–348, 2025.
- [11] X. Sun and P. Chang, “Robust perspective correction for real-world crack evolution tracking in image-based structural health monitoring,” *arXiv:2506.22437*, 2025.
- [12] NEC Corporation, “Photographing guide device,” US Patent 12,223,639 B2 (priority 2019), 2025.
- [13] S. Liao, Z. Mo, Y. Hu, and S. Z. Li, “Open-set person re-identification,” *arXiv preprint arXiv:1408.0872*, 2014.
- [14] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
- [15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.

- [16] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.
- [17] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8922–8931.
- [18] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.
- [19] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers and distillation through attention," in *International Conference on Machine Learning (ICML)*, 2021, pp. 10 347–10 357.
- [20] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [21] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.
- [22] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *IEEE/CVF Int. Conference on Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.
- [23] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.
- [24] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," <https://github.com/ultralytics/ultralytics>, 2023.
- [25] L. Maiano, A. Montuschi, M. Caserio *et al.*, "A deep-learning-based antifraud system for car-insurance claims," *Expert Systems with Applications*, vol. 231, p. 120644, 2023.
- [26] M. J. Hasan, C. K. Nguyen, Y. L. Boo, H. Jahani, and K.-L. Ong, "Vehicle damage detection using artificial intelligence: A systematic literature review," *WIREs Data Mining and Knowledge Discovery*, vol. 15, p. e70027, 2025.
- [27] S. A. Pérez-Zarate, D. Corzo-García, J. L. Pro-Martín *et al.*, "Automated car damage assessment using computer vision: Insurance company use case," *Applied Sciences*, vol. 14, no. 20, p. 9560, 2024.
- [28] D. Mallios, L. Xiaofei, N. McLaughlin, J. M. Del Rincon, C. Galbraith, and R. Garland, "Vehicle damage severity estimation for insurance operations using in-the-wild mobile images," *IEEE Access*, vol. 11, pp. 78 644–78 655, 2023.
- [29] K. Sakurada and T. Okatani, "Change detection from a street image pair using CNN features and superpixel segmentation," in *British Machine Vision Conference (BMVC)*, 2015.
- [30] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Autonomous Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
- [31] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [32] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," in *Readings in Computer Vision*. Morgan Kaufmann, 1987, pp. 726–740.
- [33] D. Barath, J. Matas, and J. Noskova, "MAGSAC: Marginalizing sample consensus," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 197–10 205.
- [34] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1858–1865, 2008.
- [35] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.